

SMART ACCIDENT DETECTION & EMERGENCY ALERT SYSTEM

Mini Project Documentation

Domain: Electronics & Communication Engineering (ECE)

Development Platform: Base44 AI

Application Type: Responsive Web Application

Live Project: <https://tasteful-safe-drive-alert.base44.app>

1. Project Overview

The Smart Accident Detection & Emergency Alert System is an ECE mini project designed to improve emergency response after road accidents. The proposed system combines accident detection, location information, emergency notification concepts and a web-based monitoring dashboard. The web application was developed using Base44 AI as an AI-assisted application development platform.

2. Objectives

- Detect possible vehicle accident conditions.
- Provide a quick emergency alert concept.
- Display vehicle and accident information through a web dashboard.
- Provide accident location using GPS information.
- Maintain accident/event history.
- Manage emergency contact information.
- Demonstrate AI-assisted application development for an ECE project.

3. Problem Statement

Road accidents can result in serious injuries or loss of life when emergency assistance is delayed. A victim may not be able to contact family members, emergency services may not receive information immediately, and the exact accident location may be difficult to communicate. Therefore, a smart system is proposed to detect an accident condition and provide relevant emergency information quickly.

4. Proposed Solution

The proposed system combines sensor-based accident detection with a web-based monitoring dashboard. The overall concept is: Vehicle → Sensors → Microcontroller → Accident Detection → Location Data → Emergency Alert → Web Dashboard. For demonstration, simulation data may be used when physical sensors or an ESP32 are not connected.

5. System Architecture

Vehicle/Sensors → ESP32 or Controller → Accident Detection Logic → Emergency Alert and Web Application → User/Emergency Contact.

6. Working Principle

1. Sensors monitor acceleration, gyroscope movement, speed, temperature and battery condition.
2. The controller checks for abnormal impact or movement.
3. GPS can provide latitude and longitude.
4. When an accident is confirmed, an emergency event is generated.
5. The web dashboard displays the accident status and relevant information.
6. Accident/event information can be stored for later reference.

7. Web Application Features

Dashboard: Safe/Warning/Accident status, vehicle information, GPS, speed, impact level and battery status.

Accident Detection: Displays the current accident condition.

Emergency Alert: Shows accident time, location, impact level and contact information.

GPS Location: Displays latitude, longitude and map information.

Sensor Monitoring: Displays accelerometer, gyroscope, speed, temperature, battery and impact data.

Accident History: Maintains previous event records.

Emergency Contacts: Allows contact information to be maintained.

8. Simulation Mode

The prototype can include Safe, Warning and Accident simulation conditions. In Accident mode, abnormal sensor values are generated, the dashboard changes to an accident state, location information is displayed, an accident record is created and emergency contact information is shown. Simulation data should be clearly identified as demo data.

9. Technologies Used

Hardware concepts: ESP32/Microcontroller, accelerometer, gyroscope, GPS module, GSM/Wi-Fi communication, buzzer and battery.

Software/Web: Base44 AI, web application, HTML/CSS/JavaScript concepts, responsive UI, dashboard and data storage.

10. Role of Base44 AI

Base44 AI was used as the primary application development platform. Natural-language requirements can be used to create and refine the web application interface, workflows, data structures and deployment. The platform supports AI-assisted application development and related backend capabilities.

11. Application Flow

START → Monitor Vehicle → Read Sensor Data → Check Abnormal Impact → Verify Accident → Get Location → Generate Alert → Update Dashboard → Save Event → END.

12. Advantages

- Faster accident notification concept
- Location-based emergency information
- Easy-to-use dashboard
- Real-time monitoring concept
- Accident history management
- Expandable IoT architecture
- Suitable for ECE applications

13. Limitations

- The current prototype may use simulated sensor data.
- Physical sensor integration is required for real-world detection.

- GPS accuracy depends on environmental conditions.
- Network connectivity may be required.
- Accident-detection algorithms require extensive testing before practical deployment.

14. Future Scope

- Real ESP32 integration
- Real-time GPS transmission
- GSM emergency SMS
- AI/ML-based accident classification
- Reliable emergency calling integration
- Secure cloud database
- Multiple-vehicle monitoring
- Mobile application
- Advanced accident analytics

15. Applications

Potential applications include private vehicles, school/college buses, public transportation, logistics vehicles, fleet management and smart transportation systems.

16. Project Outcome

The project demonstrates how an ECE-based safety concept can be combined with AI-assisted web application development. It provides a foundation for integrating embedded hardware, sensors, GPS, communication systems and a web dashboard into an accident-monitoring solution.

17. Live Project

Project URL: <https://tasteful-safe-drive-alert.base44.app>

18. Project Information

Project Title: Smart Accident Detection & Emergency Alert System

Project Type: ECE Mini Project

Development Platform: Base44 AI

Domain: ECE / IoT / Embedded Systems / Web Application

19. Conclusion

The Smart Accident Detection & Emergency Alert System is a practical ECE mini-project concept aimed at reducing delays in accident notification. By combining sensor-based accident detection, GPS location, emergency alert concepts, IoT and a web-based monitoring dashboard, the project demonstrates a potential approach to improving vehicle safety. The Base44 AI platform enables rapid development of the web application and provides a foundation for future integration with real hardware and advanced accident-detection algorithms.

References

Base44 AI App Builder: <https://base44.com/ai-app-builder>

Base44 Developers: <https://base44.com/developers>

Base44 Documentation: <https://docs.base44.com/>